

# Simulation of extreme values with Generative Adversarial Networks

Edinburgh Summer School on Generative AI for Extremes

Michaël Allouche   *Stéphane Girard*   Emmanuel Gobet

Inria, Grenoble, France.

8–11 September 2026  
JCMB, Edinburgh



THE UNIVERSITY  
of EDINBURGH



# Motivations

▷ **Risk assessment:** extreme values have low probability / high impact.



credit: edelit



credit: latimes

▷ Estimation of **extreme risk measures** (such as  $q(\alpha)$  the quantile / VaR / return level) requires dedicated methods: when  $\alpha \rightarrow 1$ ,

$$\mathbb{P}(q(\alpha) > \max_{i=1, \dots, n} X_i) = \mathbb{P}(q(\alpha) > X_1)^n = \alpha^n = e^{-n(1-\alpha)(1+o(1))} \rightarrow 1$$

if  $n(1 - \alpha) \rightarrow 0$ . The quantile is almost surely asymptotically outside the observation range  $\{X_1, \dots, X_n\}$ .

# Two lines of work

## – Rare event type simulation methods –

- ▷ Existing simulation methods (Importance Sampling, MCMC) work well but high complexity cost.
- ▷ Focus on new **data-based generative models** trained on
  - true data sets (*i.e.* historical financial returns),
  - outputs of black-box models (*i.e.* meteorological models).
- ▷ **EV-GAN**: [Allouche et al., 2022]
- ▷ **ExceedGAN**: [Allouche et al., 2026]

## – Extrapolation based on Extreme-Value Theory (EVT) –

- ▷ **Higher order** extrapolation methods leveraging the approximation properties of **neural networks** [Allouche et al., 2024].

# Simulation methods: Outline

- 1 **Generative models:** GANs & their limitations.
- 2 First method: **EV-GAN**.
- 3 Second method: **ExceedGAN**.
- 4 **Numerical experiments:** simulated & real data.
- 5 **Conclusion.**

# 1. Generative models

Let  $X$  denote a r.v. taking values in  $\mathbb{R}^d$  from which we have observations  $\{X_1, \dots, X_n\}$ . The problem is to find a function  $G : \mathbb{R}^{d'} \rightarrow \mathbb{R}^d$  (**generator**) and a (**latent**) probability distribution  $\pi$  on  $\mathbb{R}^{d'}$  such that

$$X \stackrel{d}{=} G(Z) \text{ and } Z \sim \pi.$$

**Existence:** Kuratowski's theorem (measurable isomorphism theorem)

[Villani, 2009, Page 7]

**Questions:** How to choose  $\pi$  (and  $d'$ )? How to build  $G$ ?

**Estimate  $G$ :** Generative Adversarial Networks (GANs) [Goodfellow et al., 2014]

# Generative Adversarial Networks (GANs)

**Data:**  $\{X_1, \dots, X_n\}$  iid from some unknown density  $p_X$  on  $\mathbb{R}^d$ .

## Generator:

- Given a parametric set of functions  $\mathcal{G} = \{G_\theta : \mathbb{R}^{d'} \rightarrow \mathbb{R}^d, \theta \in \Theta\}$ ,
- estimate parameters  $\theta^*$  s.t.  $\tilde{X} := G_{\theta^*}(Z) \stackrel{d}{\simeq} X$ .

## Discriminator:

- Given a parametric set of functions  $\mathcal{D} = \{D_\phi : \mathbb{R}^d \rightarrow [0, 1], \phi \in \Phi\}$ ,
- estimate the probability  $D_{\phi^*}(x)$  that  $x$  is drawn from  $p_X$ .

## Loss function:

$$\min_{\theta \in \Theta} \max_{\phi \in \Phi} \mathbb{E}_{p_X}(\log D_\phi(X)) + \mathbb{E}_{p_Z}(\log(1 - D_\phi(G_\theta(Z))))$$

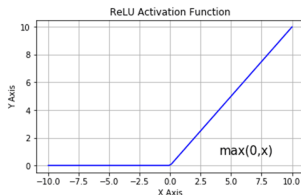
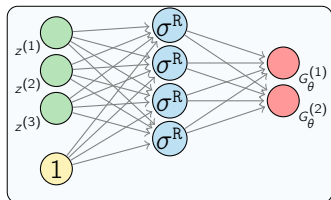
# GANs: Generator

## Neural network implementation:

$$G_{\theta}(\cdot) = \left( G_{\theta}^{(1)}(\cdot), \dots, G_{\theta}^{(d)}(\cdot) \right)$$

$$G_{\theta}^{(m)} \left( z^{(1)}, \dots, z^{(d')} \right) = \sum_{j=1}^J a_j^{(m)} \sigma^{\text{R}} \left( \sum_{i=1}^{d'} w_j^{(i)} z^{(i)} + b_j \right) + c^{(m)},$$

for  $m \in \{1, \dots, d\}$  with  $J$  neurons and ReLU (Rectified Linear Unit) function  $\sigma^{\text{R}}(x) := \max(0, x)$ .



$$d' = 3, \quad d = 2, \quad J = 4 \quad \text{and} \quad c^{(1)} = c^{(2)} = 0.$$

# GANs: Statement of the problem

Consider the simple case  $d = 1$ .

- Let  $X$  be a rv with continuous and strictly increasing cdf  $F$ .
- $X \stackrel{d}{=} q(Z)$  where  $Z \sim U([0, 1]) =: \pi$  (so that  $d' = 1$ ) and  $q(\cdot) = F^{-1}(\cdot) =: G(\cdot)$  is the quantile function.
- The generator approximates  $q(z)$ ,  $z \in (0, 1)$  by

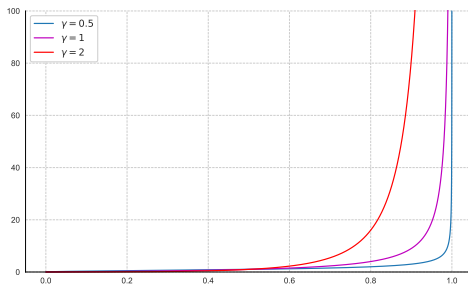
$$G_{\theta}(z) = \sum_{j=1}^J a_j \sigma^R(w_j z + b_j) + c.$$

- The Universal approximation theorem [Cybenko, 1989] only holds for **continuous functions** on a **compact set**.



# GANs: Statement of the problem

- Quantile function of a Burr distribution (heavy-tailed distribution)  
 $u \mapsto q(u)$  with tail parameter  $\gamma \in \{1/2, 1, 2\}$ .



- If  $Z$  is uniform, then  $\tilde{X} := G_{\theta}(Z)$  is bounded.
- If  $Z$  is Gaussian, then  $\tilde{X} := G_{\theta}(Z)$  is light-tailed.

# GANs: Solutions proposed in the literature

- Use a light-tail model for  $Z$  on transformed data and then transform back the generator outputs for recovering the heavy-tail property:
  - Lambert fct  $W(x) = x \exp(\gamma x^2/2) \rightarrow$  **Quant-GAN** [Wiese et al., 2020],
  - cdf of the **EVD**( $\gamma$ )  $\rightarrow$  **evtGAN** [Boulaguiem et al., 2022].*In both cases, the tail-index  $\gamma$  has to be estimated first.*
- Use a heavy-tailed distribution for  $Z$ :
  - Student-t [Feder et al., 2020],
  - **GPD**( $\gamma$ )  $\rightarrow$  **Pareto-GAN** [Huster et al., 2021].*In both cases, the tail-index  $\gamma$  has to be estimated first.*
- EVT on margins, GAN on dependence  $\rightarrow$  **WA-GAN** [Lhaut et al., 2026].
- New cost function: Exploit the joint elicibility property of the VaR and ES risk measures  $\rightarrow$  **Tail-GAN** [Cont et al., 2022].
- Heuristics:  $\rightarrow$  **Ex-GAN** [Bhatia et al., 2020].

## Other architectures.

- Variational Auto Encoders (VAE) [Lafon et al., 2023],
- $d$ -max-decreasing neural networks [Hasan et al., 2022].

# EV-GAN: Leveraging extreme-value theory

(**H**<sub>1</sub>): The cdf of  $X \geq 1$  is supposed to be **heavy-tailed**, i.e. the survival function  $\bar{F} := 1 - F$  is **regularly-varying** with index  $-1/\gamma$ ,

$$\bar{F}(x) = x^{-1/\gamma} L(x), \quad x > 0,$$

(notation:  $\bar{F} \in \mathcal{RV}_{-1/\gamma}$ ), where:

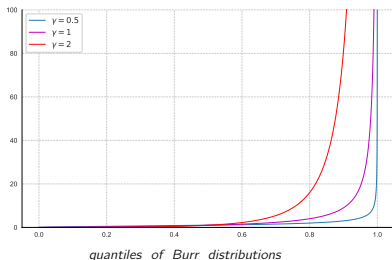
- $\gamma > 0$  is called the **tail-index**;
- $L$  is a slowly-varying function (notation:  $L \in \mathcal{RV}_0$ );

$$\frac{L(\lambda x)}{L(x)} \rightarrow 1, \text{ as } x \rightarrow \infty \text{ for all } \lambda > 0.$$

- $F$  is said to belong to the Fréchet Maximum domain of attraction.
- As a consequence of [de Haan and Ferreira, 2006, Proposition B.1.9.9], the tail quantile function  $t > 1 \mapsto U(t) := q(1 - 1/t) \in \mathcal{RV}_\gamma$ .

# EV-GAN: Tail-index function (TIF)

Since  $q(1 - 1/\cdot) \in \mathcal{RV}_\gamma$ , then for all  $u \in (0, 1)$ ,  $q(u) = (1 - u)^{-\gamma} L\left(\frac{1}{1-u}\right)$ , with  $L \in \mathcal{RV}_0$ .



Introduce, for all  $u \in (0, 1)$ :

$$f^{\text{TIF}}(u) := -\frac{\log q(u)}{\log\left(\frac{1-u^2}{2}\right)}$$

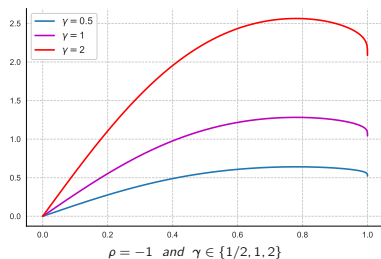
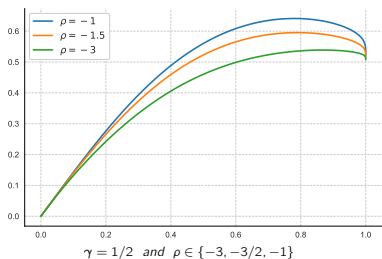
- ▷ Taking the log yields  $\log q(u) \sim -\gamma \log(1 - u)$  as  $u \rightarrow 1$ , since  $\log L(v)/\log v \rightarrow 0$  as  $v \rightarrow \infty$  [Bingham et al., 1987, Proposition 1.3.6];
- ▷ Note also that  $\log((1 - u^2)/2) \sim \log(1 - u)$  as  $u \rightarrow 1$ ;
- ▷ As a conclusion:  $f^{\text{TIF}}(u) \rightarrow \gamma$  as  $u \rightarrow 1$ .

# EV-GAN: Tail-index function (TIF)

## Proposition

Under  $(\mathbf{H}_1)$ ,  $f^{\text{TIF}}$  is a **continuous** and **bounded** function on  $[0, 1]$ . Besides,  $f^{\text{TIF}}(0) = 0$  and  $f^{\text{TIF}}(u) \rightarrow \gamma$  as  $u \rightarrow 1$ .

*Tail-index functions associated with Burr distributions*



- The Universal approximation theorem applies to the TIF.
- Is it possible to get higher regularity properties?

# EV-GAN: Second-order condition

**Second order condition.** There exist  $\rho < 0$  and  $|A| \in \mathcal{RV}_\rho$  s.t.

$$\frac{1}{A(t)} \log \left( \frac{L(zt)}{L(t)} \right) \rightarrow \frac{z^\rho - 1}{\rho}, \quad \text{as } t \rightarrow \infty, \forall z \geq 1.$$

The second-order parameter  $\rho$  tunes the rate of convergence of  $L(zt)/L(t)$  to 1.

(H<sub>2</sub>):  $L$  satisfies the second-order condition and additional regularity assumptions.

As a consequence, one can derive asymptotic expansions of the quantile  $q(u) = (1-u)^{-\gamma} L\left(\frac{1}{1-u}\right)$  and TIF functions in the neighbourhood of  $u = 1$ .

# EV-GAN: Corrected TIF (CTIF)

Introduce for all  $u \in (0, 1)$ :

$$f^{\text{CTIF}}(u) := f^{\text{TIF}}(u) - \sum_{k=1}^6 \kappa_k e_k(u),$$

where

- $\{\kappa_1, \dots, \kappa_6\}$  are coefficients depending on the distribution  $F$ ,
- $\{e_1, \dots, e_6\}$  are fixed functions independent from  $F$ .

This correction allows us to get higher regularity provided  $\rho$  is not too large.

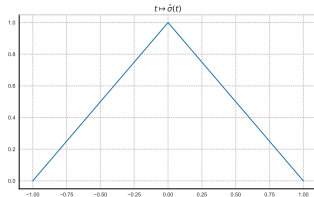
## Proposition

- (i) If  $(\mathbf{H}_1)$  holds, then  $f^{\text{CTIF}} \in \mathcal{C}^0([0, 1])$ .
- (ii) If, moreover,  $(\mathbf{H}_2)$  holds with  $\rho < -1$ , then  $f^{\text{CTIF}} \in \mathcal{C}^1([0, 1])$ .
- (iii) If, moreover,  $\rho < -2$ , then  $f^{\text{CTIF}} \in \mathcal{C}^2([0, 1])$ .

# EV-GAN: Approx. of a $\mathcal{C}^2$ function with ReLUs

**Triangular function.** Built with 3 shifted ReLUs  $\sigma^R(x) := \max(0, x)$ .

$$\hat{\sigma}^R(t) := \sigma^R(t+1) - 2\sigma^R(t) + \sigma^R(t-1)$$



**Piecewise linear approximation.** Let  $M > 0$ ,  $\delta = 1/M$  and  $t_j = j\delta$  for  $j = 0, \dots, M$ . If  $f \in \mathcal{C}^2([0, 1])$ , then

$$\sup_{t \in [0, 1]} \left| f(t) - \sum_{j=0}^M f(t_j) \hat{\sigma}^R\left(\frac{t - t_j}{\delta}\right) \right| = \mathcal{O}(1/M^2).$$

- Approximation of order  $1/M^2$  with  $J = 3(M+1)$  ReLUs.
- Similar result for  $\mathcal{C}^1$  functions.



# EV-GAN: Main result

Applying the previous construction to the CTIF yields:

## Theorem

Assume  $(\mathbf{H}_1)$  and  $(\mathbf{H}_2)$  hold. For all  $m \in \{1, \dots, d\}$ , consider the (modified) generator:

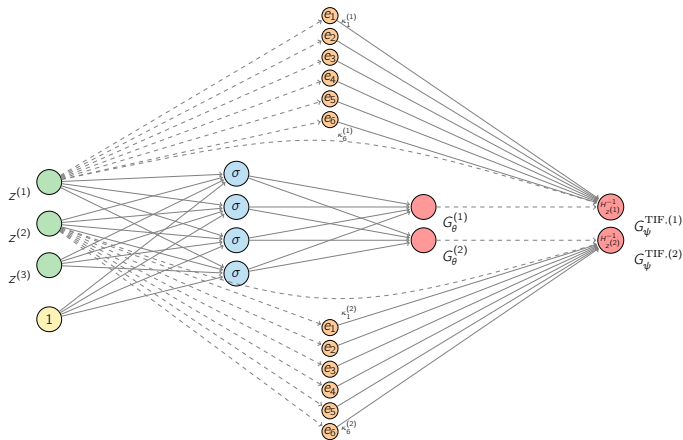
$$\tilde{G}_{\psi}^{(m)}(z) = \sum_{j=1}^J a_j^{(m)} \sigma^{\mathbf{R}} \left( \sum_{i=1}^{d'} w_j^{(i)} z^{(i)} + b_j \right) + \sum_{k=1}^6 \kappa_k^{(m)} e_k \left( z^{(m)} \right),$$

where  $\psi = \left( a_j^{(m)}, b_j, w_j^{(i)}, \kappa_k^{(m)} \right)_{i,j,k,m}$ . If  $\rho^{(m)} < -2$  then

$$\inf_{\psi} \max_{m \in \{1, \dots, d\}} \sup_{z \in [0,1]^{d'}} \left| f^{\text{TIF},(m)}(z^{(m)}) - \tilde{G}_{\psi}^{(m)}(z) \right| = \mathcal{O}(1/J^2).$$

**Remark.** Similar result when  $-2 \leq \rho < -1$  with reduced rate of convergence.

# EV-GAN: Architecture of the generator ( $d' = 3$ , $d = 2$ , $J = 4$ neurons)



——→ Trainable parameter

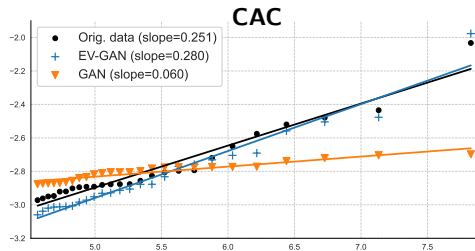
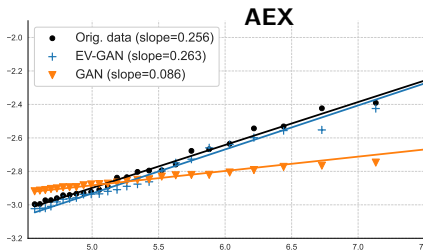
-----→ Non-trainable parameter with fixed value 1

# EV-GAN: Illustration on European stock indices

## Setting.

- Log-returns on selected heavy-tailed financial stock market indices.
- Processing of missing data: compute and synchronize log-returns.

Margins fit: log quantile-quantile plot  $-\log(1-u) \mapsto \log q(u)$  for  $u \geq 0.99$ .



# EV-GAN: Illustration on world stock indices

## Performance assessment.

- Marginals: Mean Squared Logarithmic Error

$$\text{MSLE}(\xi) = \frac{1}{d[(1-\xi)n]} \sum_{j=1}^d \sum_{i=1}^{\lceil (1-\xi)n \rceil} \left( \frac{\log(X_{n-i+1,n}^{(j)}) - \log(\tilde{X}_{n-i+1,n}^{(j)})}{\log(2)} \right)^2,$$

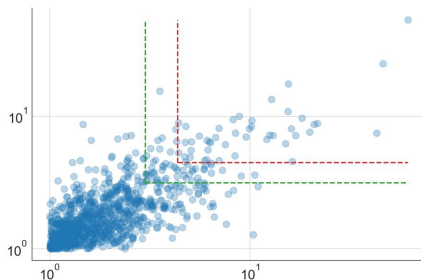
- Dependence: Absolute Kendall Error (AKE).

| ticker          | NKX   |       | Europe |       | Asia  |       | World  |       |
|-----------------|-------|-------|--------|-------|-------|-------|--------|-------|
| dimension $d$   | 1     |       | 2      |       | 3     |       | 6      |       |
| sample size $n$ | 3173  |       | 2504   |       | 1378  |       | 548    |       |
| MSLE(0.90)      | 0.010 | 0.003 | 0.080  | 0.003 | 0.049 | 0.014 | 0.069  | 0.018 |
| MSLE(0.95)      | 0.015 | 0.002 | 0.103  | 0.004 | 0.031 | 0.012 | 0.093  | 0.025 |
| MSLE(0.99)      | 0.029 | 0.004 | 0.058  | 0.007 | 0.022 | 0.011 | 0.104  | 0.041 |
| AKE             | —     | —     | 16.807 | 4.697 | 9.760 | 4.872 | 24.781 | 3.533 |

*Performance comparison between the best GAN (left) and EV-GAN (right).*

### 3. ExceedGAN: Motivations

- Given an independent sample  $\{X_1, \dots, X_n\}$  from a **multivariate** random vector  $X$ ;
- Our goal is to build a GAN to sample new data points (called **exceedances**) from  $X \mid X > u$  (componentwisely) where  $u$  is a **high multivariate** threshold.



*Illustration for two bivariate thresholds  $u$ .*

# ExceedGAN: Simplified framework

- Let us consider first the **univariate** setting.
- The high threshold  $u$  is chosen as a quantile of  $X$ :  $u = q_X(1 - \delta)$ .
- As already seen, to build a GAN for  $Y = X \mid X > q_X(1 - \delta)$ , one can use as **generator**  $G = q_Y$  and **latent distribution**  $\pi = U([0, 1])$ .
- There is a simple link between the quantiles of  $Y$  and  $X$ . For any  $z \in (0, 1)$ :

$$q_Y(1 - z) = q_X(1 - \delta z).$$

- As explained at the very beginning, if  $n\delta z$  is small, then  $q_X(1 - \delta z)$  is likely to be an extreme quantile **beyond the observation range**. Estimating the generator  $G = q_Y$  is then an **extrapolation problem**.

# ExceedGAN: Extrapolation principle

**Goal:** To link the quantity of interest  $q_Y(1-z) = q_X(1-\delta z) = U(1/(\delta z))$ , with the threshold  $q_X(1-\delta) = U(1/\delta)$ .

Under  $(\mathbf{H}_1)$ , recall that  $U(t) = t^\gamma L(t)$  with  $L \in \mathcal{RV}_0$ .

**Idea.** Introduce the log-spacing function,

$$(x_1, x_2 > 0) \mapsto \varphi(x_1, x_2) := \log \left( \frac{L(\exp(x_1 + x_2))}{L(\exp(x_2))} \right)$$

so that

$$\begin{aligned} \log U(1/(\delta z)) - \log U(1/\delta) &= \gamma \log(1/z) + \varphi(\log(1/z), \log(1/\delta)) \\ q_Y(1-z) &= q_X(1-\delta) z^{-\gamma} \exp(\varphi(\log(1/z), \log(1/\delta))) \end{aligned}$$

## Unknown quantities.

- 1 Intermediate quantile  $U(1/\delta)$
- 2 Tail index  $\gamma$
- 3 Log-spacing function  $\varphi(\cdot, \cdot)$

## Weissman. [Weissman, 1978]

- 1  $X_{n-k+1,n}$ ,  $k = \lfloor n\delta \rfloor$
- 2  $\hat{\gamma}(k)$  [Hill, 1975]
- 3 0

# ExceedGAN: Bias correction (second order)

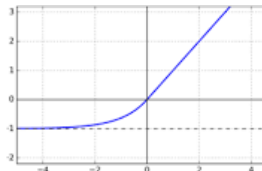
Recall the **second order condition**: There exist  $\rho < 0$  and  $|A| \in \mathcal{RV}_\rho$  s.t.

$$\frac{1}{A(t)} \log \left( \frac{L(zt)}{L(t)} \right) \rightarrow \frac{z^\rho - 1}{\rho}, \quad \text{as } t \rightarrow \infty, \forall z \geq 1.$$

Assuming  $A(t) = ct^\rho$  yields a parametric approximation of  $\varphi(x_1, x_2)$  as

$$\begin{aligned} \tilde{\varphi}(x_1, x_2) &= c \exp(\rho x_2) (\exp(\rho x_1) - 1) / \rho \\ &= c (\sigma^E(\rho(x_1 + x_2)) - \sigma^E(\rho x_2)) / \rho, \end{aligned}$$

where  $\sigma^E(x) = x\mathbb{I}\{x \geq 0\} + (\exp(x) - 1)\mathbb{I}\{x < 0\}$  is the **eLU** function.



$$\begin{cases} x & x \geq 0 \\ \alpha(e^x - 1) & x < 0 \end{cases}$$

$\Rightarrow$  **Approximation of the log-spacing function using only 2 neurons!**



# ExceedGAN: Bias correction ( $J$ -th order)

**$J$ -th order condition.** There exist  $\forall j \in \{2, \dots, J\}$ ,  $\rho_j \leq 0$  and functions  $A_j(t) \rightarrow 0$  as  $t \rightarrow \infty$  s.t. for all  $y \geq 1$ :

$$\log \left( \frac{L(yt)}{L(t)} \right) = \sum_{j=2}^J \prod_{\ell=2}^j A_\ell(t) R_j(z) + o \left( \prod_{j=2}^J A_j(t) \right) \quad \text{as } t \rightarrow \infty,$$

$$\text{where } R_j(z) = \int_1^y y_2^{\rho_2-1} \int_1^{y_2} y_3^{\rho_3-1} \dots \int_1^{y_{j-1}} y_j^{\rho_j-1} dy_j \dots dy_3 dy_2.$$

Proposition (Approximation with  $J(J-1)$  neurons)

Assume the  $J$ -th order condition holds with  $A_j(t) = c_j t^{\rho_j}$ , where  $c_j \neq 0$  and  $\rho_j < 0$  for  $j \in \{2, \dots, J\}$ . Then, for all  $x_1 > 0$  and  $x_2 > 0$

$$\varphi(x_1, x_2) = \underbrace{\sum_{i=1}^{J(J-1)/2} w_i^{(1)} \left( \sigma^E \left( w_i^{(2)} x_1 + w_i^{(3)} x_2 \right) - \sigma^E(w_i^{(4)} x_2) \right)}_{\tilde{\varphi}(x_1, x_2; \theta)} + o(\dots)$$

with  $w_i^{(1)} \in \mathbb{R}$ ,  $w_i^{(2)} < 0$ ,  $w_i^{(3)} < 0$ ,  $w_i^{(4)} < 0$ ,  $\forall i \in \{1, \dots, J(J-1)/2\}$ .

# ExceedGAN: Theoretical result

## Neural Network approximation.

$$\tilde{q}_Y(1 - z; w_0, \theta) := q_X(1 - \delta_n) z^{-w_0} \exp(\tilde{\varphi}(\log(1/z), \log(1/\delta_n); \theta))$$

where  $\theta = (w_i^{(1)}, w_i^{(2)}, w_i^{(3)}, w_i^{(4)})_{1 \leq i \leq J(J-1)/2}$ .

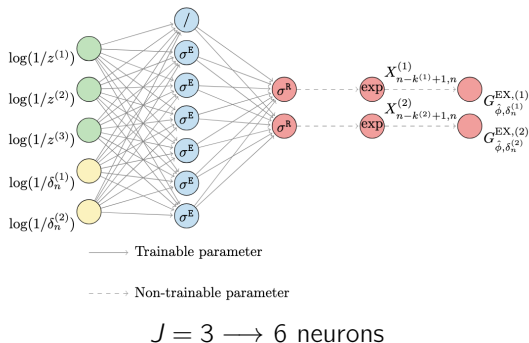
### Theorem

Assume the conditions of the above proposition hold. Then, there exists a parameter  $(w_0^*, \theta^*)$  and a threshold  $t_0 \in (0, 1)$  such that the **one hidden-layer** NN with  $J(J-1)$  neurons verifies

$$\sup_{z \in (0,1]} |\log q_Y(1 - z) - \log \tilde{q}_Y(1 - z; w_0^*, \theta^*)| \leq |\bar{\rho}_J \bar{c}_J| \delta_n^{-\bar{\rho}_J},$$

for all  $0 < \delta_n \leq t_0$ , and where  $\bar{c}_J = c_2 \times \cdots \times c_J$ ,  $\bar{\rho}_J = \rho_2 + \cdots + \rho_J$ .

# ExceedGAN: Architect. of the generator ( $d' = 3, d = 2$ )



## Two versions:

- 1 **Fixed-level (FL):** Plug the ExceedGAN generator into the GAN optimization problem for a fixed level  $\delta_n = (\delta_n^{(1)}, \dots, \delta_n^{(m)})^\top$ .
- 2 **Level-varying (LV):** Conditional extension on  $\delta_n \in \Delta_n$  with adapted optimization problem and learning algorithm, see [Allouche et al., 2026, Section 3.2]

# ExceedGAN: Validation on simulations

**Data:** Exceedances are computed in the (bivariate) upper orthant

$$Q(\delta_n) := \left\{ \mathbf{x} \in \mathbb{R}^2 \mid x^{(1)} > F_{X^{(1)}}(1 - \delta_n^{(1)}), x^{(2)} > F_{X^{(2)}}(1 - \delta_n^{(2)}) \right\}$$

from  $n = 100,000$  synthetic data from a Gumbel copula (dependence parameter  $\mu \in \{1.1, 2, 10\}$ ) and Burr margins  $(\gamma, \rho^{(1)})$ ,  $(\gamma, \rho^{(2)})$  where  $\gamma \in \{0.1, 0.5, 0.9\}$  and  $(\rho^{(1)}, \rho^{(2)}) \in \{(-1, -2), (-1, -3), (-2, -3)\}$  using a basic acceptance-rejection method.

## GAN, EV-GAN, FL-ExceedGAN.

- ①  $\delta_n = (0.1, 0.1)^\top$  with  $n/100 = 1,000$  training and 10,000 testing points.
- ②  $\delta_n = (0.05, 0.05)^\top$  with  $n/400 = 250$  training and 10,000 testing points.

## LV-ExceedGAN.

- Adaptation of the GAN algorithm to learn  $Y = X \mid X > q_X(1 - \delta_n)$  **conditionally** on  $\delta_n \in [0, 0.5]^2$ .
- $n/4 = 25,000$  training and 10,000 testing points.

# ExceedGAN: Validation on simulations

MSLE ( $\delta_n = (0.1, 0.1)^\top$ )

|          |                    | $\mu = 1.1$ |             |             |             | $\mu = 2$ |        |             |             | $\mu = 10$ |        |             |             |
|----------|--------------------|-------------|-------------|-------------|-------------|-----------|--------|-------------|-------------|------------|--------|-------------|-------------|
| $\gamma$ | $(\rho_1, \rho_2)$ | GAN         | EV-GAN      | ExceedGAN   |             | GAN       | EV-GAN | ExceedGAN   |             | GAN        | EV-GAN | ExceedGAN   |             |
| 0.3      | $(-1, -2)$         | 0.46        | 0.52        | <b>0.30</b> | 0.38        | 0.23      | 0.26   | <b>0.11</b> | 0.46        | 0.34       | 0.97   | 0.18        | <b>0.16</b> |
|          | $(-1, -3)$         | 0.98        | 0.36        | <b>0.35</b> | 3.28        | 0.43      | 0.24   | <b>0.13</b> | 0.31        | 0.44       | 0.84   | 0.22        | <b>0.20</b> |
|          | $(-2, -3)$         | 0.53        | 0.35        | 0.39        | <b>0.24</b> | 0.14      | 0.24   | <b>0.11</b> | 0.76        | 0.29       | 1.29   | <b>0.05</b> | 0.12        |
| 0.5      | $(-1, -2)$         | 4.59        | 1.49        | 1.07        | <b>1.03</b> | 3.18      | 1.11   | <b>0.54</b> | 6.72        | 2.64       | 1.33   | <b>0.39</b> | 1.01        |
|          | $(-1, -3)$         | 3.93        | 1.21        | 1.19        | <b>1.16</b> | 3.98      | 1.41   | 1.30        | <b>1.19</b> | 2.38       | 1.50   | <b>0.22</b> | 2.60        |
|          | $(-2, -3)$         | 6.03        | <b>1.09</b> | 1.67        | 2.45        | 1.22      | 3.05   | <b>0.69</b> | 2.53        | 1.93       | 1.30   | <b>0.51</b> | 2.09        |
| 0.9      | $(-1, -2)$         | 61.92       | <b>6.13</b> | 6.68        | 13.54       | 35.82     | 3.77   | <b>2.11</b> | 15.72       | 28.88      | 4.83   | <b>1.32</b> | 86.05       |
|          | $(-1, -3)$         | 38.65       | <b>4.50</b> | 4.85        | 25.65       | 44.17     | 3.82   | <b>2.59</b> | 13.13       | 24.56      | 5.69   | <b>1.20</b> | 84.16       |
|          | $(-2, -3)$         | 41.19       | 5.49        | <b>5.05</b> | 21.92       | 45.03     | 4.11   | <b>1.86</b> | 15.15       | 36.71      | 3.73   | <b>1.56</b> | 83.37       |

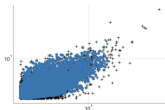
MSLE ( $\delta_n = (0.05, 0.05)^\top$ )

|          |                    | $\mu = 1.1$ |        |              |              | $\mu = 2$ |        |             |             | $\mu = 10$ |        |             |             |
|----------|--------------------|-------------|--------|--------------|--------------|-----------|--------|-------------|-------------|------------|--------|-------------|-------------|
| $\gamma$ | $(\rho_1, \rho_2)$ | GAN         | EV-GAN | ExceedGAN    |              | GAN       | EV-GAN | ExceedGAN   |             | GAN        | EV-GAN | ExceedGAN   |             |
| 0.3      | $(-1, -2)$         | 2.98        | 1.38   | 1.14         | <b>0.36</b>  | 2.10      | 0.86   | 1.91        | <b>0.36</b> | 1.23       | 1.90   | 1.22        | <b>0.62</b> |
|          | $(-1, -3)$         | 2.75        | 2.76   | <b>0.56</b>  | 2.60         | 2.35      | 1.28   | 1.50        | <b>0.89</b> | 1.35       | 1.82   | 1.27        | <b>0.34</b> |
|          | $(-2, -3)$         | 2.34        | 1.12   | 2.10         | <b>0.17</b>  | 1.88      | 1.39   | 1.47        | <b>0.46</b> | 0.75       | 1.88   | 1.19        | <b>0.43</b> |
| 0.5      | $(-1, -2)$         | 18.58       | 6.19   | 2.21         | <b>1.18</b>  | 13.88     | 5.22   | <b>2.21</b> | 6.64        | 8.87       | 3.76   | 2.77        | <b>1.60</b> |
|          | $(-1, -3)$         | 15.96       | 4.71   | 5.55         | <b>1.57</b>  | 15.08     | 3.20   | <b>2.17</b> | 2.40        | 9.06       | 4.37   | <b>2.64</b> | 3.16        |
|          | $(-2, -3)$         | 20.91       | 7.16   | 5.59         | <b>1.11</b>  | 14.23     | 4.08   | <b>2.16</b> | 3.93        | 6.24       | 5.04   | <b>2.18</b> | 5.28        |
| 0.9      | $(-1, -2)$         | —           | 16.38  | 13.70        | <b>12.97</b> | —         | 9.27   | <b>7.81</b> | 15.72       | 90.41      | 15.66  | <b>7.10</b> | 88.78       |
|          | $(-1, -3)$         | —           | 22.34  | 17.54        | <b>14.21</b> | —         | 12.06  | <b>6.82</b> | 12.91       | —          | 15.62  | <b>7.37</b> | 86.84       |
|          | $(-2, -3)$         | —           | 20.21  | <b>14.31</b> | 18.13        | —         | 11.90  | <b>9.96</b> | 20.70       | —          | 34.13  | <b>7.46</b> | 82.66       |

$$\delta_n = (0.1, 0.1)^\top, \mu = 2, (\rho_1, \rho_2) = (-1, -3)$$

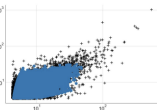
GAN

$\gamma = 0.3$



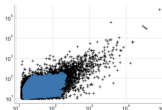
(a)

$\gamma = 0.5$



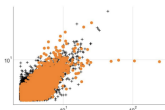
(b)

$\gamma = 0.9$

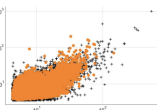


(c)

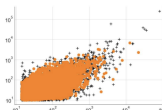
EV-GAN



(d)

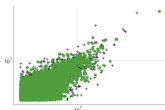


(e)



(f)

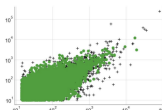
FL-ExceedGAN



(g)

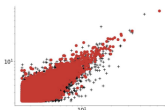


(h)

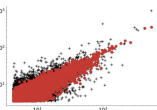


(i)

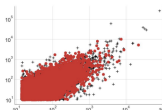
LV-ExceedGAN



(j)



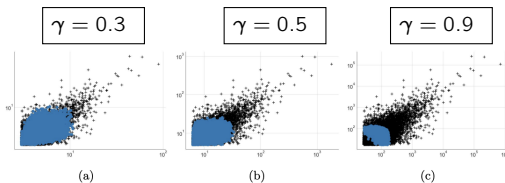
(k)



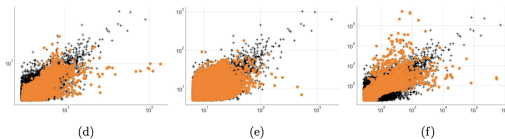
(l)

$$\delta_n = (0.05, 0.05)^\top, \mu = 2, (\rho_1, \rho_2) = (-1, -3)$$

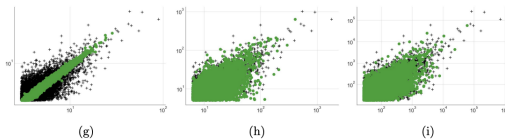
GAN



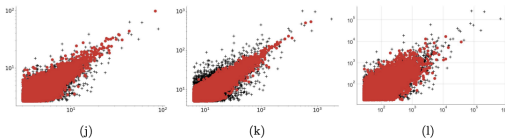
EV-GAN



FL-ExceedGAN



LV-ExceedGAN



## 4. Conclusion

- Usual GAN: not able to reproduce heavy-tailed events, leading to a severe **under-estimation of extreme risks**.
- EV-GAN: new parametrization of a **generative model** dedicated to **extreme events**.
- ExceedGAN: Improve EV-GAN results on margins using (refined) **extrapolation methods**.
- Further work: Combine the results of [Girard et al., 2026] (showing mathematically how the **extreme dependence structure** can be preserved) with the previous ones on the margins.



# References I



Allouche, M., Girard, S., and Gobet, E. (2022).  
EV-GAN: Simulation of extreme events with ReLU neural networks.  
*Journal of Machine Learning Research*, 23(150):1–39.



Allouche, M., Girard, S., and Gobet, E. (2024).  
Estimation of extreme quantiles from heavy-tailed distributions with neural networks.  
*Statistics and Computing*, 34:12.



Allouche, M., Girard, S., and Gobet, E. (2026+).  
ExceedGAN: simulation above extreme thresholds using Generative Adversarial Networks.  
*Extremes*.



Bhatia, S., Jain, A., and Hooi, B. (2020).  
ExGAN: Adversarial generation of extreme samples.  
*arXiv preprint arXiv:2009.08454*.

# References II



Bingham, N. H., Goldie, C. M., and Teugels, J. L. (1987).  
*Regular variation*, volume 27 of *Encyclopedia of Mathematics and its Applications*.  
Cambridge University Press, Cambridge.



Boulaguiem, Y., Zscheischler, J., Vignotto, E., van der Wiel, K., and Engelke, S. (2022).  
Modeling and simulating spatial extremes by combining extreme value theory with generative adversarial networks.  
*Environmental Data Science*, 1:e5.



Cont, R., Cucuringu, M., Xu, R., and Zhang, C. (2022).  
Tail-GAN: Nonparametric scenario generation for tail risk estimation.  
arXiv preprint [arXiv:2203.01664](https://arxiv.org/abs/2203.01664).



Cybenko, G. (1989).  
Approximation by superpositions of a sigmoidal function.  
*Mathematics of Control, Signals, and Systems*, 2(4):303–314.

# References III



de Haan, L. and Ferreira, A. (2006).

*Extreme value theory.*

Springer Series in Operations Research and Financial Engineering. Springer, New York.



Feder, R. M., Berger, P., and Stein, G. (2020).

Nonlinear 3D cosmic web simulation with heavy-tailed generative adversarial networks.

*Physical Review D*, 102(10):103504, 18.



Girard, S., Gobet, E., and Pachebat, J. (2026+).

HTGAN: Heavy-tail GAN for multivariate dependent extremes via latent-dimensional control.

*International Journal of Computer Mathematics.*

# References IV



Goodfellow, I., Pouget-Abadie, J., Mirza, M., Xu, B., Warde-Farley, D., Ozair, S., Courville, A., and Bengio, Y. (2014).

Generative adversarial nets.

In *Advances in neural information processing systems*, pages 2672–2680.



Hasan, A., Elkhailil, K., Ng, Y., Pereira, J. M., Farsiu, S., Blanchet, J. H., and Tarokh, V. (2022).

Modeling extremes with  $d$ -max-decreasing neural networks.

arXiv preprint [arXiv:2102.09042](https://arxiv.org/abs/2102.09042).



Hill, B. M. (1975).

A simple general approach to inference about the tail of a distribution.

*The Annals of Statistics*, 3(5):1163–1174.



Huster, T., Cohen, J. E., Lin, Z., Chan, K., Kamhoua, C., Leslie, N., Chiang, C. J., and Sekar, V. (2021).

Pareto GAN: Extending the representational power of GANs to heavy-tailed distributions.

*arXiv preprint arXiv:2101.09113.*



Lafon, N., Naveau, P., and Fablet, R. (2023).

A VAE approach to sample multivariate extremes.

*arXiv preprint arXiv:2306.10987.*



Lhaut, S., Rootzén, H., and Segers, J. (2026+).

Simulation of multivariate extremes: A Wasserstein–Aitchison GAN approach.

*Extremes.*



Villani, C. (2009).

*Optimal transport*, volume 338 of *Fundamental Principles of Mathematical Sciences*.

Springer-Verlag, Berlin.



Weissman, I. (1978).

Estimation of parameters and large quantiles based on the  $k$  largest observations.

*Journal of the American Statistical Association*, 73(364):812–815.



Wiese, M., Knobloch, R., Korn, R., and Kretschmer, P. (2020).

Quant GANs: deep generation of financial time series.

*Quantitative Finance*, 20(9):1419–1440.